



# Executive Field Summary: Prototype #1

Rampur, Assam — 4 m³ Fixed-Dome Biodigester — Operational since Jan 15, 2026

438 kg

CO<sub>2</sub>e AVOIDED / MO

600 kg

WASTE DIVERTED / MO

120 kg

FIREWOOD REPLACED / MO

5.26 t

CO<sub>2</sub>e TARGET / YEAR

3 hrs

SMOKE-FREE / DAY



## Environmental

### Stewardship

Capturing methane from 600 kg of waste avoids **218 kg CO<sub>2</sub>e/month**. Displacing 120 kg of firewood avoids **220.1 kg CO<sub>2</sub>e/month**. Combined: **438.1 kg CO<sub>2</sub>e/month** (IPCC Tier-1, AR6 GWP).



## Social Capacity

**5 jobs** created locally. **4 operators** trained (exceeding target). **3 hrs/day** smoke-free indoor air, reducing PM2.5 exposure. **1 company** registered.



## Financial

### Transparency

**€3,000** Catalyst Grant + **€600** Private Equity. 100% allocated locally: construction (€1,559), IT (€952), labor (€303), transport (€115).

### Monthly CO<sub>2</sub> Emissions Avoided or a rural biodigester project

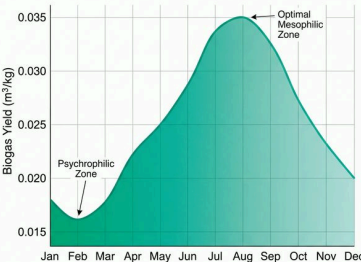


**Total: 438.1 kg CO<sub>2</sub>e/month avoided**

Monthly CO<sub>2</sub>e Breakdown

### Biogas Yield vs. Temperature

A biodigester locator in Assam, India



Biogas Yield vs Temperature

### Grant Utilization (€3,000)



+ €600 Private Equity Co-financing

Grant Utilization (€3,000)



## Advanced Biological Modeling

The dashboard dynamically adjusts biogas projections using Assam's mesophilic temperature curve. Winter yield drops to 0.016 m³/kg; summer peaks at 0.035 m³/kg — reflecting real bioprocess behavior with ±15% organic variance applied weekly.